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DESIGN AND IMPLEMENTATION OF FIR FILTER BASED ON FPGA

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Abstract— In this study, a FIR (Finite Impulse Response) filter which is one of numerical filters was dealt with in 8th grade and realized on Field Programmable Gate Array (FPGA) platform. FIR filter design was done with MATLAB platform. Vivado software has been used as the development environment and the filter was implemented on the Nexys 4 FPGA platform using the VHDL programming language. With the help of the designed filter, different noise signals are applied to the FPGA input and the desired noiseless signal is obtained on the output.

Keywords— FIR Filter, FPGA, Filter design, FPGA based FIR filter

I. INTRODUCTION

Filters are components of analogue or digital form, which are frequently used in electronics and, as the name implies, perform filtering on specific criteria on signals. They are used to obtain the desired signal by removing unwanted signals from noisy signals. In short, they can be defined as signal selection system.

Digital filters provide more flexibility compared to analogue filters, allowing the design to be easily modified using software. They are also less sensitive to changes in external parameters such as temperature, magnetic declination. Digital processing of a signal also reduces processing costs. With that, digital data storage is easy.

Filtering is very important in many designs. Filters can be used to extract the desired signal to be accessed, which is buried under too much noise. Furthermore, whether a system is linear or not, the output can be determined by filtering at certain frequencies. These features serve to identify systems.

Many different calculation techniques are used in the filtering process. One of them is the convolution process is performed by multiplying each coefficient in the equation by multiplying the values of the signal. The quick realization of these calculations is very important for real-time applications.

For example, it is necessary to use fast and high precision filters to separate the ECG signals used in the diagnosis of heart diseases from other external ambient noise.

Many DSP applications require a high degree of FIR filter. However, the computational complexity increases with the increase in the filter level due to the necessity of larger calculations. This is a serious challenge in real-time implementation of the filter [1].

The FPGA is able to perform the calculations on the signals quickly and simultaneously with the parallel processing capability. Thus, almost real time filtering can be carried out on FPGA.

II. FILTERS

Filtering is very important in many designs. Filters can be used to extract the desired signal to be accessed, which is buried under too much noise. Furthermore, whether a system is linear or not, the output can be determined by filtering at specific frequencies. These features serve to identify systems.

A. Filter Types

Filters can be in different ways. For example, there may be fluctuations in the transition bands or they may have a flat transition band. Filters are typically categorized by shape in the formats specified in the list:

1. Low-pass filter: It is the filter type that passes the signals below a certain frequency.
2. Band-pass filter: It is the type of filter that stops the signals outside the upper and lower frequency limits and passes the signals between these limits.
3. Band-stop filter: It is the type of filter that passes the signals outside the lower and upper frequency limits and stops the signals between these limits.
4. High-pass filter: It is the type of filter that passes the signals above a certain frequency.
5. All-pass filter: It is the type of filter that passes all the frequencies evenly but changes the phase relationship between the various frequencies.

B. Filter Shapes

Filters can be in different shapes. For example, there may be fluctuations in the transition bands or they may have a flat transition band. Filters are typically categorized by shape in the formats specified in the list:

1. Bessel: It has the smoothest group delay compared to other filter shapes.
2. Butterworth: It is a form of filter designed to have a frequency response as straight as possible in the transition band. Also called maximum flat amplitude.
3. Chebyshev: It is designed to have a minimum error between the ideal filter and the actual filter. They are

categorized into two types: the transition band and the fluctuation band in the stop band.

4. Elliptic: It is a form of filter with fluctuation in the transition band and stop band. But it has the fastest transition between the pass and stop band.

Choosing the filter shape to be used depends on the desired properties in the application. For example, in the output band, the output signal amplitude is needed to be able to track the input signal amplitude at the possible level. In this case, a Butterworth filter can be used to provide more passband in the system. When the output signal frequency is desired to follow the input signal exactly with a linear phase response, Bessel filter shape should be used due to its structure.

C. Analogue and Digital Filters

In another aspect, the filters can be formed in two ways as analogue and digital filters. Passive filters in an analogue circuitry are the steps of inductors, capacitors or resistors. Active analogue filters can be in a structure that uses amplifiers or resonators. The value can be determined using already created tables or applications to design analogue filters.

D. FIR Filter

Another filter application of digital filters is the FIR (finite pulse response) filter. FIR filters do not use feedback and their outputs are only connected with existing and previous inputs. The output signal does not supply its own input. FIR filters are always stable because they are only dependent on inputs. Another digital filter type, IIR, takes input from its output and is therefore a filter that uses feedback. FIR filters need a higher order to meet the same specifications as IIR (endless pulse response) filters.

FIR filters consist of three main components: unit delay element, multiplication and summation elements. The number of the unit delay element means the end of the pulse response time. In addition, the number of unit delay elements, expressed as $n-1$, determines the degree of the filter. The multiplication elements serve to multiply the input signal with the corresponding filter coefficient. Summation elements collect the data obtained from the multiplication output and produce the filter output. This process is called convolution. The filter coefficient used determines the degree of the filter. The quality of the filter increases as the filter grade increases [2]. From this information, the equation of an FIR filter is;

$$y(n) = \sum_{k=0}^{N-1} h_k x(n-k)$$

expressed as. According to the formula, $y(n)$ refers to the output signal, h_k frequency response coefficients and $x(n)$ input signal [3].

III. FPGA

FPGAs are semiconductor devices based on an optional adaptive logic blocks matrix connected by programmable connections. FPGAs can be reprogrammed according to the desired application or functionality requirements. Unlike other types of processors, they are capable of parallel processing and

can execute multiple operations at the same time. Generally, they are high performance devices due to their structure and are often used in high speed applications. Aviation and defense industry, data centers, video applications, medical, image processing, wireless communication are available.

Unlike other processors, connections to FPGAs are not fixed, they can be customized. The main advantage of the FPGA solution is that when a specific hardware is open to design and an error is detected in the system or when the schedule is obsolete, it is possible to change its structure, to improve and improve its functioning functions. The FPGA configuration is usually specified using a hardware description language (HDL). The most common languages are VHDL and Verilog.

The FPGA development boards are ready-to-use development environments with an FPGA processor and the required hardware. They include all the components required to program and operate an FPGA. Users are presented as a method for prototyping the systems they will develop. Unlike general-purpose systems such as a computer, the development cards usually contain no hardware or very few hardware connected to the user interface. In order to be able to install a user-developed program, they can include various ports such as serial port, flash memory port, USB port.

IV. FIR FILTER DESIGN

The FIR filter design consists of two parts: the approach to the problem and the realization of the problem. It gives the transfer function through four steps by taking the approach stage features [4]. These stages:

1. In the frequency domain, the desired or ideal response is selected.
2. Select a permitted class of filters (N for FIR filters).
3. Select a measure of approach quality.
4. Choose a method or algorithm to find the best filter transfer function.

The realization step is to select the structure to implement the transfer function, which may be in the form of a circuit diagram or in the form of a program. There are basically three ideal methods for FIR filter design:

1. Windowing method,
2. Impact response cutting method,
3. Optimal filter design method.

The method of cutting the pulse response is the simplest design method. It has undesirable frequency field characteristics and is not very useful. The window method is simple and useful but not optimal [5].

The FIR filter is designed using the window method. Filter coefficients were found by this method. The main idea of the window method is to have the narrowest main lobe and as little side lobes as possible. The Hamming window filter design method is the method that provides the specified properties [6]. The window method calculates $h_d(n)$ with the inverse Fourier transform based on the ideal desired filter frequency response. Since $h_d(n)$ is long, the window function must be used to reach the unit impulse response $h(n)$ that meets the requirement. The calculation equation is shown below.

$$h(n) = w(n) * h_d(n)$$

Where $w(n)$ is the window function [7]. The coefficients of the FIR filter are obtained using the Discrete Fourier Transform (DFT) of the required frequency transfer function by a known windowing method. The window method performs the mathematical function whose values are zero outside the selected range. Compared to other design methods, the window method is very easy to design and implement [8].

For design and simulation, the filter design application of the Matlab software, FDATool (Filter Design and Analysis Tool) was used. In the design, the filter was set at 8th degree, the sampling rate was determined as $f_s=96000$ and the cut frequency $f_c=10800$. The Matlab application interface used in the design is given in Fig.1.

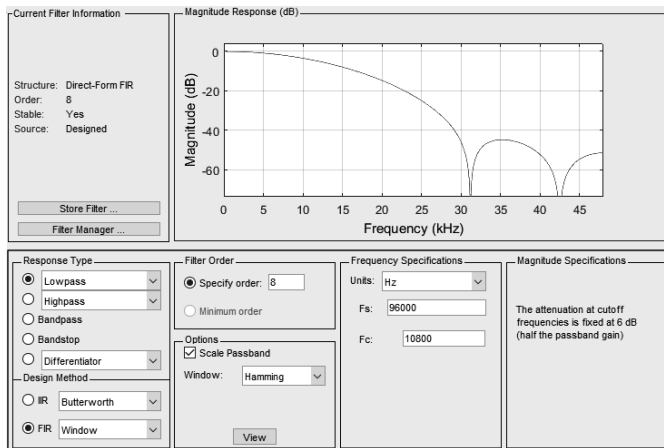


Fig.1 FIR Filter Design with FDATool

The coefficients of the filter designed via FDATool are exported to a txt file and prepared for VHDL implementation. The coefficients calculated with FDATool are given in Table 1.

TABLE I
FIR Filter Coefficient

0.002473596173554889
0.024426175761756149
0.10673334784122872
0.22491170669296029
0.28291034706099982
0.22491170669296029
0.10673334784122872
0.024426175761756149
0.002473596173554889

These coefficients are defined as variables in the VHDL software and are used to calculate the output signal by using the input signal applied. Design architecture is in parallel structure. The filter should be developed by developing a signal flow diagram describing the sample sequences in terms of operations. Direct Form I uses a $2N$ delay unit to implement the transfer function with N according to the order of the filter. This layout is suitable for low-level filters, but not practical for higher-level filters. Direct Form II potentially needs only N delay elements that are half of the Direct Form I scheme [8]. In this study,

Direct Form II scheme was used in the structure of the filter applied.

V. APPLICATION AND RESULTS

FPGA is a powerful medium that supports hardware and parallel operation for testing, analyzing and designing a hardware or software. VHDL (very fast hardware definition language) also provides the support and the ability to describe the best design and design.

An exemplary signal is generated in the Matlab environment for the simulation of the FIR filter. This signal was randomly generated by adding a noise signal to the signal. The results are given in Fig.2.

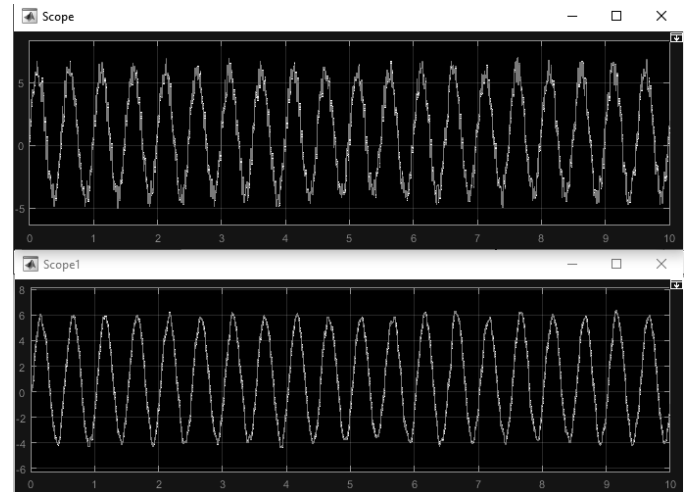


Fig.2 Noisy Signal / Filtered Signal

As seen in Figure 2, the noise on the filter which is designed is eliminated. The higher the degree of the filter, the closer the output signal is to the desired value.

The VHDL language was used to define the FIR filter to the hardware. The application was carried out on Xilinx's Vivado FPGA software and performed on the Nexys 4 FPGA card. The prepared coefficients with the Matlab software using FDATool were converted to integer values for ease of calculation and defined as constant value variables in VHDL language. As a result of the calculations, the output was obtained by re-arranging at the ratio converted to an integer. Port entry for filter input and filter output has been made for implementation on FPGA. In the VHDL language of the FIR filter, memory variables for a clock frequency, filter grade and convolution process are defined.

For the convolution process of the filter, the input signal must be kept in memory and the memory must be same with the filter length. The input values for convolution are saved by shifting to memory array.

Fig. 3 shows an input signal applied to the VHDL simulation and the stored data.

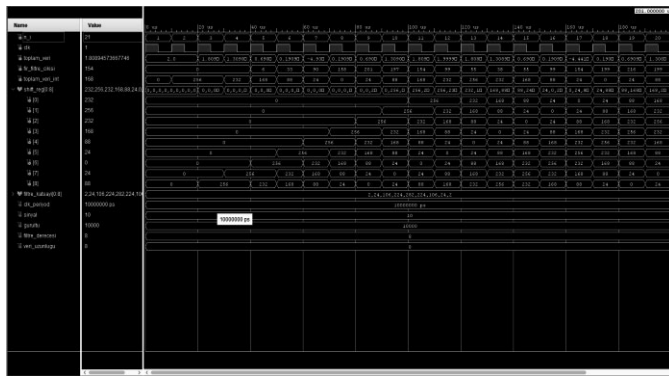


Fig.3 VHDL Simulation / Convolution

In order to obtain the output signal, the stored input signal is multiplied by the filter coefficients and each product is collected. After convolution, the signal is filtered and the desired signal is obtained. In practice, a 10 Hz noise was added to a 10 KHz signal, and 10 KHz noiseless signal was obtained at the filter output.

Fig.4 shows the simulation results of the signal filtered and separated from the noise.

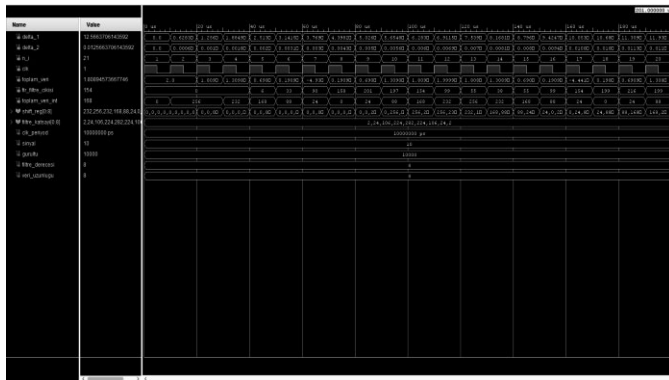


Fig.3 VHDL Simulation / Filtered Signal

As a result of the application, an 8th degree FIR filter was obtained. The application of the FIR filter on the FPGA provides a speed advantage in the process due to the ability of the FPGA performing parallel operation. This allows the use of a higher degree of filters.

Increasing the degree of filter requires many mathematical operations to achieve the desired result due to the convolution process used in the calculation. FPGA performs each mathematical operation in separate blocks at the same time and enables us to obtain the result. Therefore, it increases the filter coefficient and enables us to reach the desired signal faster and more noise-free.

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